英文回答

自我介绍（自）

CN：各位老师，早上好！我是焦元，来自甘肃敦煌，今年 22 岁，即将毕业于成都理工大学环境科学与工程专业，很荣幸参加本次复试。我将从个人情况、科研经历和未来规划三方面介绍自己。

我喜欢运动、阅读科幻和历史书籍，热衷计算机科学，熟练使用 Linux，能写 Shell、Python 和前端代码，对 AI 兴趣浓厚，常用 DeepSeek、Grok 等工具，并部署过 Qwen 模型，运营博客分享经验。

科研上，我专注生物信息学。大二参与生命科学竞赛，研究 OsSGL2 基因对高温胁迫的影响，负责系统发育和蛋白质分析，获国家级二等奖并发表综述。大三主持黑麦草重金属胁迫基因分析，完成 RNA-seq 和 WGCNA 研究。今年 1 月起，在四川科伦博泰实习，开发 AI 转化医学平台，部署 Alphafold3，用 Vue 和 Flask 设计应用，并开发数据库与可视化工具。

未来，我希望成为大学教授，硕士阶段努力适应科研，争取硕博连读，或扎实完成学业，为博士打基础。我期待在老师指导下，深耕生物信息学与 AI 的交叉应用。谢谢！

EN：Good morning, teachers! I’m Jiao Yuan from Dunhuang, Gansu, 22 years old, and about to graduate from Chengdu University of Technology with a degree in Environmental Science and Engineering. It’s an honor to participate in this graduate interview. I’ll introduce myself in three aspects: personal background, research experience, and future plans.

I enjoy sports, reading sci-fi and history books, and am passionate about computer science. I’m skilled in Linux, can code in Shell, Python, and front-end languages, and have a strong interest in AI. I frequently use tools like DeepSeek and Grok, have deployed the Qwen model, and run a blog to share my insights.

In research, I focus on bioinformatics. As a sophomore, I joined the 2022 Life Science Competition, studying the impact of OsSGL2 gene overexpression on heat stress resistance, handling phylogenetic and protein interaction analyses. The project won a national second prize and resulted in a published review. In my junior year, I led a study on ryegrass genes under heavy metal stress, conducting RNA-seq and WGCNA analyses. Since January this year, I’ve interned at Sichuan Kelun-Biotech, developing an AI-driven translational medicine platform, deploying Alphafold3, building Vue and Flask applications, and creating databases with visualization tools.

Looking ahead, I aspire to become a university professor. During my master’s, I’ll adapt to research demands, aiming for a combined master’s-PhD program if possible, or completing my master’s diligently to prepare for a PhD. I look forward to exploring the intersection of bioinformatics and AI under your guidance. Thank you!

自我介绍（模板）

CN：尊敬的教授们，早上好！感谢给我这次面试的机会。我叫焦元，今年 22_岁，来自甘肃敦煌。我目前是成都理工大学环境科学与工程的一名大四学生。进入大学后，我一直认真对待自己的专业课，努力掌握了一定的专业知识。大二的时候，我由于参加生命科学竞赛被生物信息学深深吸引，于是选择跨专业考研。我认为攻读生物信息学硕士学位也可以充分发挥我学科背景的优势。接下来，我想谈谈我的未来计划。在接下来的三年里，我计划阅读大量相关文献，形成对报考专业特别是生物信息学研究方向的更系统看法。除此之外，我将专注于我的专业领域的学习，努力提高我的研究能力。我的自我介绍完毕。如果我有幸被录取，我将在未来三年内全身心地投入到学习和研究中。

EN：Good morning, dear professors! It’s my great honor to be here for this interview. My name is Jiao Yuan, 22 years old, and I come from Dunhuang, Gansu. I am currently a senior student in Environmental Science and Engineering at Chengdu University of Technology. After entering the university, I have always taken my specialized courses seriously and worked hard to master certain knowledge about my major. In my sophomore year, I was attracted to bioinformatics through participating in a life science competition, so I chose to switch my major in postgraduate study. Besides, I believe that pursuing a master's degree in bioinformatics can give full play to the advantages of my disciplinary background. Next, I want to talk about my future plans. In the next three years, I plan to read vast relevant literature to form a more systematic view of bioinformatics, especially in the research direction of bioinformatics. Besides that, I will concentrate on the study of my professional field and work hard to improve my research ability. That’s all about me. If I am lucky enough to be admitted, I will devote myself to the study and research in the next three years.

科研经历

CN：2022 年 12 月至 2023 年 12 月，我作为成都理工大学黄进课题组成员，协助完成植物抗逆相关课题，负责 RNA-seq、基因家族分析等工作，参与发表 4 篇论文，主要负责图表绘制与系统发育分析，并搭建和运营课题组官网；在此期间，我还作为项目负责人参与了黑麦草重金属助迫响应基因的筛选与生物信息学分析项目，负责 RNA-seq、WGCNA 分析及基因家族鉴定，并参加了 2023 年第九届全国大学生生命科学竞赛（科学探究类），同时参与了过表达 OSSGL2 基因对植物高温助迫抗性影响的项目，负责基因家族分析及论文撰写，获得 2022 年第八届全国大学生生命科学竞赛（科学探究类）国家级二等奖。

2024 年 12 月至 2025 年 1 月，我在山东省农业科学院畜牧兽医研究所担任实习生，负责 ONT 三代全长转录组与全基因组甲基化分析，并进行 SNP 鉴定与等位基因分析；2025 年 1 月至 2025 年 6 月，我在四川科伦博泰生物医药公司担任生物信息实习生，使用 Vue + Flask 开发科伦转化医学 AI 应用平台，部署 DeepSeek、Alphafold3 等模型，并开发医学数据库及配套的可视化应用。

EN：From December 2022 to December 2023, I served as a member of Professor Huang Jin's research group at Chengdu University of Technology, where I assisted in completing projects related to plant stress resistance. My responsibilities included conducting RNA-seq and gene family analysis, and I contributed to the publication of four papers, primarily focusing on chart creation and phylogenetic analysis. Additionally, I built and maintained the official website for the research group. During this period, I also led a project on screening and bioinformatics analysis of heavy metal stress response genes in ryegrass, where I was responsible for RNA-seq, WGCNA analysis, and gene family identification. I participated in the 9th National College Student Life Science Competition (Scientific Inquiry Category) in 2023. Concurrently, I was involved in a project investigating the effects of overexpressing the OSSGL2 gene on plant heat stress resistance, where I conducted gene family analysis and contributed to paper writing, earning a National Second Prize in the 8th National College Student Life Science Competition (Scientific Inquiry Category) in 2022.

From December 2024 to January 2025, I worked as an intern at the Shandong Academy of Agricultural Sciences, Institute of Animal Science and Veterinary Medicine, where I was responsible for ONT full-length transcriptome and whole-genome methylation analysis, as well as SNP identification and allele analysis. From January 2025 to June 2025, I served as a bioinformatics intern at Sichuan Kelun-Biotech Pharmaceutical Company, where I developed an AI application platform for translational medicine using Vue + Flask, deployed models such as DeepSeek and Alphafold3, and created a medical database along with its visualization tools.

毕业设计

CN：论文题为《基于机器学习的华西雨屏区土壤容重空间分布特征研究》，属于环境科学与工程专业。华西雨屏区是我国重要生态屏障，但面临水土流失和土地退化问题。土壤容重是评估土壤质量的关键，影响植被生长和水分渗透。传统测量方法费时且覆盖范围有限。为此，我结合机器学习、遥感数据和环境变量，研究该区土壤容重分布及其驱动因素，为生态恢复提供科学依据。

我采集了 24 个站点的土壤容重数据，整合降雨、地形、植被等环境因子，构建数据库。使用随机森林和 XGBoost 模型，筛选关键变量并预测分布特征。当前，我正分析海拔、坡度、降雨对土壤容重的影响，以揭示分布规律。研究建立了土壤容重与环境因子的数据库，明确了主要驱动因素。结果将支持土壤质量评估和生态修复。创新点在于首次应用机器学习实现大范围高效预测。

EN：The thesis, titled "Study on the Spatial Distribution Characteristics of Soil Bulk Density in the Rain-Shelter Area of Western China Based on Machine Learning," pertains to the field of Environmental Science and Engineering. The rain-shelter area in western China is a vital ecological barrier in the country, yet it faces significant challenges such as soil erosion and land degradation. Soil bulk density, a critical indicator for evaluating soil quality, affects vegetation growth and water infiltration. Traditional measurement approaches are time-intensive and limited in spatial scope. To overcome these limitations, I combined machine learning techniques with remote sensing data and environmental variables to explore the spatial distribution of soil bulk density and its driving factors in this region, thereby providing a scientific foundation for ecological restoration.

I gathered soil bulk density data from 24 sites and integrated environmental factors—including precipitation, topography, and vegetation—to create a comprehensive database. By employing Random Forest and XGBoost models, I identified key variables and predicted the spatial distribution patterns. Currently, I am examining the effects of altitude, slope, and precipitation on soil bulk density to uncover its distribution 规律 (patterns). The study established a database linking soil bulk density with environmental factors and elucidated the primary driving forces. These findings will facilitate soil quality assessments and ecological restoration initiatives. The innovation of this research lies in its pioneering use of machine learning to achieve efficient, large-scale predictions in this region.